Viraj Mankani

Computer Science & Engineering • Backend Systems • Distributed Infrastructure • Applied AI

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EDUCATION

National Institute of Technology, Tiruchirappalli (NIT Trichy) *Aug. 2023 – July 2027*

Bachelor of Technology in Computer Science and Engineering **CGPA: 8.10 / 10.0**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming, Distributed Systems, Software Engineering

HONORS & ACADEMIC ACHIEVEMENTS

- **Joint Entrance Examination (JEE):** Secured **All India Rank (AIR) 950** out of 1.6 million (16 lakh) participating candidates nationwide.
- **Maharashtra Common Entrance Test (MHT-CET):** Secured **State Rank 1** across the entire state of Maharashtra.
- **Competitive Programming & Problem Solving:** Solved **500+ Data Structures & Algorithms questions** across LeetCode and GeeksforGeeks; achieved a **1303 peak rating** on Codeforces.

WORK EXPERIENCE

Tata Communications *May 2026 – July 2026*

Project Trainee, Software Engineering (Backend & Cloud Infrastructure) • *Certificate* *Mumbai, India*

- **High-Throughput Enterprise Backend Microservices:** Architected an asynchronous core backend platform in Python and FastAPI using non-blocking event loops, dependency injection (*Depends*), and Pydantic v2 data contracts; engineered Async SQLAlchemy 2.0 connection pooling and PostgreSQL schema design across 30+ asynchronous REST APIs, powering central auth and resource coordination across 8 internal applications for 100+ daily enterprise users.
- **Distributed Event Processing, Caching & Idempotency:** Engineered an asynchronous distributed event and background task processing engine using Redis and *asyncio*, implementing distributed locks, idempotency keys, dead-letter queues, and automated exponential-backoff retries; built a Redis multi-tier cache-aside layer with selective TTL invalidation, slashing database query load by 60% and reducing p95 API response latency from 180ms to 25ms.
- **ASGI Middleware, Observability & Reliable CI/CD:** Implemented custom FastAPI ASGI middleware for distributed *X-Correlation-ID* request tracing, token bucket tenant rate-limiting, and structured JSON telemetry integrated with Prometheus and Grafana dashboards; authored zero-downtime asynchronous Alembic database migration pipelines and built a comprehensive test suite with *pytest-asyncio* achieving 92% code coverage on GitLab CI/CD pipelines.

Lyceum Development Centre *May 2025 – July 2025*

Machine Learning Intern (Applied AI, RAG & Vector Systems) • *Certificate* *Chennai, India*

- **RFP Parsing & Intelligent Document Ingestion:** Built automated regex-based data extraction pipelines in Python processing 300+ multi-page enterprise RFP documents per month into structured JSON schemas, eliminating manual proposal processing overhead by 80%.
- **Vector Database & Semantic Retrieval Infrastructure:** Architected and scaled ChromaDB vector database infrastructure holding 5M+ text embeddings, powering a LangChain RAG pipeline over historical pricing and occupancy datasets to improve predictive accuracy by 20% and reduce document processing latency 6x.
- **LLM-Powered RAG Application & Real-Time Inference:** Deployed a production FAQ search service across 1,800+ question-answer pairs using LangChain and transformer embeddings, sustaining sub-3s query latency and 95% retrieval precision, and shipped an interactive Gradio dashboard that cut client turnaround time from 3 hours to 30 minutes.

TECHNICAL PROJECTS

NginZ – Distributed Cloud-Native API Gateway & Service Coordinator *March 2026*

Node.js, Express.js, TypeScript, Nginx, Docker, Redis, PostgreSQL, Prometheus, Grafana • GitHub Repository

- Architected a distributed cloud-native API gateway on Nginx and Node.js/Express, coordinating 10 Dockerized instances with automatic Redis-backed service discovery, SIGTERM graceful deregistration, and zero downtime during simulated central auth outages via local JWT validation.
- Prevented cascading service degradation by engineering a Redis-backed Token Bucket rate limiter and per-service Circuit Breakers with exponential backoff and jitter; validated 1,000 RPS sustained throughput with Prometheus metrics and 5 live Grafana dashboards.
- Built a React/TypeScript operator dashboard with live traffic simulators and fault injection tools, enabling chaos engineering experiments across network partitions and high-load traffic spikes.

Shopizer – Production-Grade Multi-Tier AWS Cloud Deployment *April 2026*

Java, Spring Boot, AWS (EC2, ALB, RDS, ECR, IAM, S3, VPC), Docker, MySQL, Maven • GitHub Repository

- Containerized an enterprise Java Spring Boot e-commerce monolith into Docker services and deployed on AWS across an Application Load Balancer (ALB) and Multi-AZ RDS MySQL, maintaining 99.9% target-group availability under continuous load tests.
- Hardened cloud security by replacing static credentials with IAM role-based EC2-to-ECR authentication, isolated VPC subnets, and layered least-privilege Security Groups, reducing attack surface exposure by 60%.
- Authored multi-stage Dockerfiles (Maven build stage → slim JRE runtime), reducing container image size by 65% and cutting build times by 35%; resolved JDK 17/25 build chain incompatibilities and tuned HikariCP connection pooling for zero bean-wiring errors.

Project Buddy – Real-Time Full-Stack Developer Collaboration Platform *September 2025*

React, TypeScript, Node.js, Express, MongoDB, WebSockets, D3.js, JWT, Docker • GitHub Repository

- Shipped an end-to-end developer collaboration platform with 12+ production features, enabling software engineers to discover projects, form distributed teams, and track milestones in real time.
- Built low-latency persistent multi-user chat and live team activity streams using native WebSockets backed by MongoDB; implemented JWT authentication with bcrypt password hashing and granular role-based access control (RBAC).
- Engineered interactive GitHub-style contribution heatmaps with D3.js and automated project milestone tracking; hardened all authentication and project routes against abuse with rate limiting via *express-rate-limit*.

Product Search System – Full-Stack E-Commerce Search Platform *March 2025*

MEAN Stack (MongoDB, Express.js, Angular, Node.js), Elasticsearch, TypeScript, Docker • GitHub Repository

- Built a full-stack e-commerce search platform on the MEAN stack, integrating Elasticsearch-powered full-text search, autocomplete, and relevance ranking with faceted multi-attribute filtering on a responsive Angular frontend.
- Structured a scalable catalog and query layer with MongoDB schemas and an Express/Node.js REST API for sub-second responses; containerized the entire application with Docker for consistent reproducible deployment.

Machine Unlearning using SISA Training (MUnL) *March 2025*

PyTorch, Scikit-learn, Streamlit, Distributed ML Workloads, Model Sharding • GitHub Repository

- Implemented SISA (Sharded, Isolated, Sliced, Aggregated) machine unlearning in PyTorch on FNMA loan data, cutting unlearning compute cost by 90% and training time by 80% with 95%+ retained model accuracy.
- Engineered distributed shard-and-slice isolation pipelines, retraining only affected model slices upon data deletion requests and serializing intermediate checkpoints across storage for full reproducibility.
- Delivered an interactive production ML dashboard using Streamlit on top of the preprocessing and inference pipeline for on-demand unlearning verification.

Safety Gear Detection using Vision Transformer (ViT) *August 2024*

Python, PyTorch, TorchVision, Model Serving, Inference Optimization • GitHub Repository

- Fine-tuned an 86M-parameter Vision Transformer on 3,000+ labeled industrial images, achieving ~94% test accuracy (12% gain over baseline CNN architectures) using transfer learning with Cross-Entropy loss and Adam optimizer.
- Optimized image preprocessing pipelines and inference tensor batching for real-time model serving, achieving sub-200ms latency per image on production GPU inference servers.
- Deployed an interactive visual evaluation dashboard to track confusion matrices and precision-recall metrics across compliance edge cases.

POSITIONS OF RESPONSIBILITY

Senior Data Analyst • Sigma – The Business Club of NIT Trichy *Aug. 2024 – May 2026*

- Delivered data-driven strategic insights across 3+ client consulting projects in diverse industries, helping organizations make high-confidence decisions based on structured analytics.
- Performed exploratory data analysis (EDA) on datasets of 100K+ records, uncovering operational trends, anomalies, and underlying root causes.
- Built and deployed predictive machine learning models achieving up to 92% accuracy, presenting 10+ actionable recommendations that boosted client operational efficiency by 25% and reduced annual operating costs by up to 10%.

Product Manager • The Product Folks - NITT *May 2025 – May 2026*

- Shipped 5+ product-led projects end-to-end, from ideation to deployment, driving 1,200 MAU through structured problem framing, rapid prototyping, and iterative user feedback loops.
- Oversaw digital strategy across LinkedIn and Instagram, refining brand tone and driving 30% higher engagement through feature-driven content and community sprints.

Coordinator, Technical Content Team • Pragyan, NIT Trichy *Oct. 2024 – May 2026*

- Served as Technical Content Coordinator for Pragyan ’25, NIT Trichy’s ISO 9001/20121 certified international techno-managerial fest, ensuring rigorous technical quality and editorial standards across all festival initiatives.
- Authored 5+ in-depth articles on emerging technology trends, distributed systems, and software engineering innovations, published across official media channels and acknowledged across global tech communities.

EXTRACURRICULAR ACTIVITIES

- **Community Outreach & Social Service:** Volunteered at NSS-NIT Trichy (National Service Scheme). Actively participated in community outreach programs visiting schools and elder care facilities to assist underprivileged individuals and spread educational awareness.
- **Technical & Cultural Quizzing:** Secured **1st Place** at the Main Quiz held during Vortex ’25, the annual Computer Science and Engineering symposium of NIT Trichy.

TECHNICAL SKILLS

Languages:	C, C++, Java, Python, TypeScript, JavaScript, SQL, Bash, HTML/CSS
Backend & APIs:	FastAPI (Async, Depends, Lifespan), Asyncio, Node.js, Express.js, Spring Boot, Async SQLAlchemy 2.0, Alembic, Pydantic v2, RESTful APIs, Microservices, WebSockets, JWT, OAuth2/OIDC, RBAC
Databases & Caching:	PostgreSQL, MySQL, Redis (Cache-Aside, Distributed Locks, Streams), ChromaDB (Vector DB, 5M+ embeddings), MongoDB, Elasticsearch
Cloud & Infrastructure:	Amazon Web Services (EC2, ALB, RDS, S3, IAM, VPC, ECR), Docker, Docker Compose, Kubernetes fundamentals, Nginx, Linux (Ubuntu, POSIX systems)
DevOps & Telemetry:	Prometheus, Grafana, Distributed Tracing (X-Correlation-ID), GitLab CI/CD, GitHub Actions, Pytest, Pytest-Asyncio, JUnit, Git, Maven
Applied AI & ML:	PyTorch, TorchVision, Vision Transformers (ViT), LangChain (Agents & Semantic RAG), Scikit-learn, SISA Machine Unlearning, NumPy, Pandas, Gradio, Streamlit